Research Paper - Execution Plan Split

so you have the project context. we are a team of 4. give a full detailed, person by person split, proper execution plan, to finish the task within 5 days.(or as soon as possible). the plan must inlcude proper things they have to do. it must be assigned in such a way that, 1 person wont be dependent on the other and they can finish the work independently. the plan for each person must incude steps, their goal,w aht all to do, anything i might have missed.
Show moreShow less

Yes. I have the project context now, including the **research direction, literature/novelty assessment, full implementation plan, and the existing 1-week sprint**.

One important change I would make to the existing sprint: **do not organize the team around sequential dependencies** such as "Person 2 creates data → Person 3 integrates it → Person 1 builds fusion → Person 4 evaluates." That creates a bottleneck.

Instead, for a **5-day emergency sprint**, all four people should work as **four semi-independent workstreams with fixed interfaces**. Each person should be able to complete their component using mock/synthetic inputs even if another person's component is unfinished.

The target should be:

> **By the end of Day 5: working three-stream prototype + reproducible synthetic dataset + fusion experiments + ablations + evaluation + figures + complete paper draft + documentation.**

The current sprint defines the core contribution as the **fusion architecture and explainable scoring**, rather than claiming novelty for gaze, mouse, or cameras individually. This is especially important because the literature review found that the individual components are already established, while no verified paper was found combining the exact three streams in the proposed architecture.

ONE_WEEK_SPRINT_PLAN Literature_Review_and_Novelty_A…

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1. THE 5-DAY TEAM STRUCTURE

Person 1 — Architecture + Research/Paper Lead

Owns:

- Overall system architecture
- Fusion model
- Explainability
- Research methodology
- Paper
- Final integration

Person 2 — Data + Simulation + Dataset Lead

Owns:

- Synthetic exam generator
- Gaze simulation
- Mouse simulation
- Environment simulation
- Dataset schema
- Data generation
- Train/validation/test split
- Data quality

Person 3 — Computer Vision / Three-Stream Extraction Lead

Owns:

- Gaze extraction
- Environment-camera/object detection
- Camera architecture
- Feature extraction
- Real-time inference demonstration
- CV technical methodology

Person 4 — Experiments + Evaluation + Reproducibility Lead

Owns:

- Baselines
- Training pipeline
- Ablations
- Metrics
- Fairness/error analysis
- Graphs/tables
- Reproducibility/README/testing

---

2. THE MOST IMPORTANT RULE: FIX THE INTERFACE ON DAY 1

This is what makes the team independent.

Everyone agrees that **every 10-second window** has this structure:

```
window_id
timestamp_start
timestamp_end

GAZE
    fixation_duration_mean
    fixation_count
    saccade_velocity_mean
    gaze_deviation
    gaze_confidence
    head_yaw
    head_pitch

INTERACTION
    cursor_velocity_mean
    cursor_velocity_std
    click_frequency
    keystroke_frequency
    idle_fraction
    tab_switch_count
    velocity_spike_ratio

ENVIRONMENT
    phone_detected
    phone_confidence
    notes_detected
    extra_person_count
    suspicious_objects_count

LABEL
    honest / cheating
    cheating_type
```

This follows the existing plan's 10-second window concept and feature design. ONE_WEEK_SPRINT_PLAN

The critical independence rule

Person 1 **does not wait for Person 2's final dataset**.

They create:

```
mock_dataset.csv
```

with the exact schema.

Person 2 later produces:

```
dataset.csv
```

with the same schema.

Person 3 can independently test its feature extraction against the schema.

Person 4 can independently build the evaluation pipeline against mock data.

Then everything plugs together.

---

3. FINAL DELIVERABLES AFTER 5 DAYS

By the end, you should have:

Technical

- [ ] Three-stream architecture
- [ ] Gaze feature extraction/simulation
- [ ] Mouse/interaction feature extraction/simulation
- [ ] Environment detection/simulation
- [ ] Common synchronized data format
- [ ] Synthetic dataset
- [ ] Early fusion baseline
- [ ] Late fusion baseline
- [ ] Attention-based proposed fusion
- [ ] Single-modality baselines
- [ ] Ablation experiments
- [ ] Evaluation metrics
- [ ] Explainable alert score
- [ ] Error analysis
- [ ] Fairness/sensitivity analysis
- [ ] Reproducible experiment script

Research

- [ ] Literature comparison table
- [ ] Novelty statement
- [ ] Research questions
- [ ] Methodology
- [ ] Experimental protocol
- [ ] Results
- [ ] Limitations
- [ ] Future work
- [ ] References

Paper

- [ ] Abstract
- [ ] Introduction
- [ ] Related Work
- [ ] Proposed Method
- [ ] Dataset

- [ ] Experiments
- [ ] Results
- [ ] Discussion
- [ ] Limitations
- [ ] Conclusion

Visuals

At minimum:

1. System architecture diagram
2. Fusion architecture diagram
3. Model comparison/AUC graph
4. Ablation graph
5. ROC/PR graph
6. Confusion matrix
7. Explainability example

---

4. DAY 1 — FOUNDATION + PARALLEL DEVELOPMENT

Everyone — 1 hour

Goal

Freeze the project definition so nobody changes the target halfway through.

Read:

- Literature/novelty assessment
- One-week sprint
- Project plan

The literature review's key conclusion must guide everything:

> **The novelty is the combination/fusion/deployment architecture, not the individual sensors.**

The review specifically recommends emphasizing a defined three-stream fusion model, synchronization, explainability, calibrated scores, and ablation studies.
Literature_Review_and_Novelty_A…

Decide immediately

```
Streams:
1. Gaze
2. Mouse/interaction
```

```
3. Environment camera

Window:
10 seconds

Task:
Honest vs suspicious/cheating

Dataset:
Synthetic

Fusion:
Early + Late + Attention

Primary metric:
AUC

Secondary:
F1
Precision
Recall
FPR
FNR
```

The existing sprint uses 200 synthetic sessions with a 60/20/20 train/validation/test split.

ONE_WEEK_SPRINT_PLAN

PERSON 1 — DAY 1

Goal

Freeze the scientific architecture and create everything that the other three people need as an interface.

Step 1 — Create repository

```
exam-proctoring/
│
├── data/
├── simulators/
├── gaze/
├── environment/
├── interaction/
├── fusion/
├── experiments/
├── results/
├── figures/
├── paper/
├── docs/
└── tests/
```

Step 2 — Create the data contract

Create:

```
docs/data_schema.md
```

Define every feature.

Example:

Stream	Feature
Gaze	fixation_duration_mean
Gaze	fixation_count
Gaze	saccade_velocity_mean
Gaze	gaze_deviation
Gaze	confidence
Gaze	head_yaw
Gaze	head_pitch
Mouse	cursor_velocity_mean
Mouse	cursor_velocity_std
Mouse	click_frequency
Mouse	keystroke_frequency
Mouse	idle_fraction
Mouse	tab_switch_count
Mouse	velocity_spike_ratio
Environment	phone_detected
Environment	phone_confidence
Environment	notes_detected
Environment	extra_person_count
Environment	suspicious_objects_count

Step 3 — Design fusion

Implement:

```
Model A: Gaze only
Model B: Mouse only
Model C: Environment only

Model D: Early Fusion

Model E: Late Fusion

Model F: Gaze + Mouse
Model G: Gaze + Environment
Model H: Mouse + Environment

Model I: Three-stream Attention Fusion
```

The proposed architecture uses stream-specific embeddings, multi-head attention, learned stream gates, and a classification head.

ONE_WEEK_SPRINT_PLAN

Step 4 — Define explainability

For every prediction:

```
JSON
```

```
{
  "alert_score": 0.82,
  "gaze_contribution": 0.31,
  "mouse_contribution": 0.38,
  "environment_contribution": 0.13,
  "reasons": [
    "High gaze deviation",
    "Unusual cursor velocity",
    "Phone detected"
  ]
}
```

Step 5 — Create mock data

Do **not wait for Person 2**.

Generate 100 fake rows using random values.

This lets Person 1 test the fusion model immediately.

Day 1 deliverables

- `data_schema.md`
- `fusion_architecture.md`
- fusion model skeleton
- mock dataset
- experiment definitions
- research questions
- contribution statement

PERSON 2 — DAY 1

Goal

Own the entire data-generation pipeline. Nobody else needs to generate data.

Step 1 — Build simulator architecture

Create:

```
simulators/
├── gaze_simulator.py
├── mouse_simulator.py
├── environment_simulator.py
├── exam_simulator.py
└── generate_dataset.py
```

Step 2 — Gaze simulator

Simulate:

Honest

- Mostly screen-centered gaze
- Normal fixation
- Normal saccades
- Occasional natural deviation

Suspicious

- Long off-screen gaze
- Repeated side glances
- Increased gaze variance
- Head turning

Step 3 — Mouse simulator

Generate:

- Cursor movement
- Cursor velocity
- Cursor acceleration
- clicks
- idle periods
- typing
- tab switches

Cheating scenarios:

```
phone
copy-paste
web search
external assistance
notes
```

The project plan already identifies cursor velocity, acceleration, click frequency, idle fraction, tab switching and typing characteristics as useful interaction features. Complete_Plan

Step 4 — Environment simulator

Generate:

```
normal:
    person = 1
    phone = 0
    notes = 0

phone cheating:
    phone = 1

notes cheating:
    notes = 1
```

```
external help:
    extra_person = 1
```

Step 5 — Dataset generator

Target:

```
200 sessions

100 honest
100 cheating

multiple cheating types
```

Step 6 — Split correctly

Do **not randomly split individual 10-second windows from the same session**.

This is important.

Instead:

```
Session-level split

120 sessions → train
40 sessions → validation
40 sessions → test
```

Otherwise windows from the same session can leak into both train and test.

Day 1 deliverables

- All simulator files
- Dataset schema implementation
- 20 test sessions
- dataset generator
- deterministic random seed

PERSON 3 — DAY 1

Goal

Build the real computer-vision side independently of the synthetic dataset.

Step 1 — Gaze

Use MediaPipe Face Mesh/iris landmarks as the fast baseline.

The project plan recommends MediaPipe as the initial lightweight gaze solution. Complete_Plan

Build:

```
gaze/
├── gaze_estimator.py
├── calibration.py
├── feature_extractor.py
└── demo.py
```

Step 2 — Implement

Input:

```
webcam frame
```

Output:

```Python
{
    "gaze_x": ...,
    "gaze_y": ...,
    "gaze_confidence": ...,
    "head_yaw": ...,
    "head_pitch": ...
}
```

Step 3 — Feature extraction

Convert raw gaze to:

```
fixation duration
fixation count
saccade velocity
gaze deviation
confidence
head yaw
head pitch
```

Step 4 — Environment detector

Use pretrained YOLO.

Detect:

```
person
phone
book
notebook
```

Do **not** spend Day 1 training a custom YOLO model.

The sprint explicitly assumes a pretrained YOLOv8-nano detector rather than training one from scratch.

ONE_WEEK_SPRINT_PLAN

Step 5 — Camera architecture document

Define:

```
Camera 1:
face-facing webcam
→ gaze
```

```
Camera 2:
side/rear three-quarter environment camera
→ phone/notes/person/environment
```

Do **not** claim this has been physically deployed.

The literature assessment specifically says the CCTV/physical-hall extension should be treated as a proposed deployment architecture unless experimentally validated. Literature_Review_and_Novelty_A…

Day 1 deliverables

- Gaze extractor
- Gaze feature extractor
- YOLO detector
- Environment feature extractor
- CV demo
- camera architecture diagram

PERSON 4 — DAY 1

Goal

Build the entire evaluation infrastructure independently.

This person should never have to wait for the ML model.

Step 1 — Metrics module

Create:

```
experiments/
├── metrics.py
├── evaluator.py
├── baselines.py
├── ablation.py
└── visualize.py
```

Step 2 — Implement metrics

Required:

```
AUC-ROC
F1
Precision
Recall
FPR
FNR
confusion matrix
```

Step 3 — Create fake prediction file

Example:

```csv
model,label,prediction
gaze_only,1,0.72
gaze_only,0,0.21
...
```

Then make the evaluation system work using this.

Step 4 — Define experiments

```
E1: Single stream

E2: Early fusion

E3: Late fusion

E4: Attention fusion

E5: Leave-one-stream-out

E6: Error analysis

E7: Fairness/sensitivity
```

Step 5 — Define final result table

```
Model | AUC | F1 | Precision | Recall | FPR | FNR
```

Day 1 deliverables

- Evaluation framework
- Metrics
- Result table template
- Ablation script skeleton
- Visualization skeleton
- experiment configuration

---

5. DAY 2 — BUILD EVERYTHING TO WORK INDEPENDENTLY

PERSON 1

Goal: Fusion implementation

Implement:

Early Fusion

```
[gaze + mouse + environment]
              ↓
```

```
      MLP
        ↓
   prediction
```

Late Fusion

```
gaze → classifier ┐
mouse → classifier ├→ average
env → classifier  ┘
```

Attention Fusion

```
gaze ──────┐
mouse ─────┼→ embeddings → attention → gating → classifier
environment┘
```

Also implement

- normalization
- missing stream handling
- model checkpoint saving
- inference function

Important

Implement it using **random/mock data first**.

At the end of Day 2, Person 1 should be able to execute:

```bash
python fusion/train.py --data mock.csv
```

without needing Person 2.

PERSON 2

Goal: Generate the complete dataset

Morning

Finish all simulators.

Afternoon

Generate:

```
200 sessions
```

with different scenarios.

Suggested distribution:

Type	Sessions
Honest	100
Phone	25
Notes	25
Copy/paste	25
External assistance	25

Add variability

Don't make cheating perfectly obvious.

For example:

Honest:

```
occasional gaze deviation
occasional mouse burst
long idle period
```

Cheating:

```
sometimes only one modality changes
sometimes two modalities change
sometimes all three change
```

This is essential.

Otherwise the attention model will simply learn:

```
phone_detected = cheating
```

and the research result becomes weak.

Day 2 output

```
data/synthetic/
    dataset.csv
    dataset.json
    metadata.json
```

and a report:

```
data_quality_report.md
```

PERSON 3

Goal: Make real CV extraction operational

Morning

Finish gaze.

Test:

```
face detected
eyes detected
iris detected
head pose
gaze coordinates
```

Afternoon

Finish YOLO environment detection.

Test:

```
phone
person
book
notebook
```

Create feature interface

Most important output:

```python
extract_gaze_features()
extract_environment_features()
```

which return exactly the schema defined by Person 1.

Also create

```
cv_demo.mp4
```

or screenshots demonstrating:

```
Face → gaze
Environment → object detection
```

These can later go into the paper.

PERSON 4

Goal: Build training/evaluation pipeline

Take the mock dataset and implement:

```
load data
↓
normalize
↓
train
↓
```

```
predict
↓
metrics
↓
save JSON
```

Create:

```
results/
    results.json
    metrics.csv
```

Important

Build the pipeline to accept:

```Bash
python run_experiments.py --dataset dataset.csv
```

rather than hard-coding anything.

That means when Person 2 finishes the real dataset, **Person 4 simply swaps the file**.

No rewrite.

---

6. DAY 3 — FIRST COMPLETE SYSTEM

This is the most important day.

By the end of Day 3, you want:

```
DATA
 ↓
FEATURES
 ↓
FUSION
 ↓
PREDICTION
 ↓
METRICS
```

working end-to-end.

PERSON 1 — DAY 3

Goal: Integrate fusion with actual dataset

Load Person 2's dataset.

Check:

```
missing values
feature ranges
class balance
normalization
```

Then train:

```
Gaze-only
Mouse-only
Environment-only
Early
Late
Attention
```

Also implement explainability

For every test sample:

```
Alert Score = 0.78

Evidence:
Gaze: medium
Mouse: high
Environment: high

Reason:
- gaze deviation elevated
- cursor velocity spike
- phone detected
```

PERSON 2 — DAY 3

Goal: Dataset quality + scenario diversity

Do **not** start another major feature.

Instead audit the dataset.

Check:

Distribution

```
honest vs cheating
cheating types
feature distributions
```

Correlations

Ask:

> Is the label almost directly encoded in one feature?

For example:

```
phone_detected = 1 → always cheating
```

If yes, add difficult examples:

```
phone-like object but not phone
looking away but honest
```

```
high mouse velocity but honest
tab switch caused by system notification
```

This creates realistic false positives.

PERSON 3 — DAY 3

Goal: Technical validation of the sensing pipeline

Run real webcam/video examples.

Document:

```
FPS
latency
detection confidence
failure cases
```

Test:

- glasses
- different lighting
- head movement
- partial face
- camera angle

For environment:

- phone near desk
- phone far away
- book
- notebook
- multiple people

Deliverable

```
cv_validation.md
```

with actual observations.

PERSON 4 — DAY 3

Goal: Run the first complete experiment

Use Person 2's dataset.

Run:

```
single modalities
early fusion
late fusion
attention
```

Generate:

```
results.csv
```

Do NOT fabricate results

The existing sprint document contains illustrative expected numbers such as AUC ≈ 0.85 for attention fusion, but those are targets/examples, not results you should copy into the paper. ONE_WEEK_SPRINT_PLAN

Your paper must use the actual outputs.

---

7. DAY 4 — EXPERIMENTS + SCIENTIFIC VALIDATION

Now stop adding unnecessary features.

Day 4 is experiment day.

PERSON 1 — DAY 4

Goal: Make the proposed method scientifically defensible

Run:

Experiment 1

```
Gaze
Mouse
Environment
```

Experiment 2

```
Early fusion
Late fusion
Attention fusion
```

Experiment 3

Leave-one-out:

```
Gaze + Mouse
Gaze + Environment
Mouse + Environment
All three
```

Experiment 4

Stream importance:

```
attention/gating weights
```

Experiment 5

Explainability examples.

Produce:

```
results/proposed_results.json
results/attention_weights.csv
```

PERSON 2 — DAY 4

Goal: Stress-test the dataset

Generate additional test sets.

Test A — Noisy gaze

Increase gaze noise.

Test B — Mouse noise

Increase random mouse movements.

Test C — Environment failure

Remove environment detections.

Test D — Single-modality cheating

Example:

```
cheating = phone

but gaze normal
mouse normal
environment detects phone
```

Test E — Silent cheating

```
no tab switch
no major mouse change
small gaze change
```

This directly addresses one of the known failure modes identified in the sprint plan: silent cheating can produce false negatives.

PERSON 3 — DAY 4

Goal: Produce technical evidence and figures

Generate:

Figure 1

Three-stream system.

Figure 2

Gaze pipeline.

Figure 3

Environment-camera pipeline.

Figure 4

Example detection frames.

Also provide:

```
FPS
latency
camera specifications
feature extraction timing
```

for the methodology.

PERSON 4 — DAY 4

Goal: Own all quantitative evaluation

Run:

Main metrics

```
AUC
F1
Precision
Recall
FPR
FNR
```

ROC

Generate ROC curves.

PR

Generate PR curves.

Confusion matrices

One for proposed model.

Ablation

Generate:

```
Gaze
Mouse
Environment
Gaze+Mouse
Gaze+Environment
Mouse+Environment
All Three
```

Error analysis

Find:

```
5 false positives
5 false negatives
```

and classify why they happened.

---

8. DAY 5 — PAPER + FINAL SYSTEM

This is not another coding day.

Day 5 is integration, writing, verification and packaging.

PERSON 1 — FULL DAY

Goal: Finish the paper

Write:

1. Abstract

~200–250 words.

2. Introduction

Explain:

```
problem
limitations of existing approaches
```

```
gap
research question
contribution
```

3. Related Work

Use the literature review.

Do NOT say:

> "Nobody has used gaze."

Do NOT say:

> "Nobody has used multiple cameras."

Do NOT say:

> "We invented multimodal proctoring."

The literature explicitly contradicts those claims. Literature_Review_and_Novelty_A…

Instead:

> "Our contribution is a three-stream fusion architecture integrating gaze, computer-interaction behavior, and an independently positioned environment camera, with synchronized evidence fusion and explainable incident scoring."

4. Methodology

Explain:

```
gaze
mouse
environment
synchronization
feature extraction
fusion
alert score
```

5. Dataset

Explain:

```
synthetic generation
sessions
cheating scenarios
features
splitting
```

6. Experiments

Describe every experiment.

7. Results

Insert actual tables.

8. Discussion

Explain:

- why fusion helps
- which stream contributes
- failure cases
- false positives
- limitations

9. Ethics

Explicitly state:

- AI score is not proof of cheating
- human review is required
- synthetic data limitation
- privacy/consent
- fairness
- no unqualified CCTV claim

The project's broader plan explicitly treats alerts as probabilistic and requires human review rather than punishment-oriented automated decisions. Complete_Plan

10. Conclusion

Short and strong.

PERSON 2 — DAY 5

Goal: Freeze and package the dataset

Do not keep changing the simulator.

Create:

```
data/
├── dataset.csv
├── metadata.json
├── feature_description.md
└── generation_config.json
```

Create dataset documentation

Document:

```
number of sessions
number of windows
class balance
cheating types
features
train/validation/test split
random seed
```

Verify

```
No NaNs
No duplicate sessions
No leakage
Correct labels
Correct feature ranges
```

Then freeze:

```
DATASET_VERSION = 1.0
```

PERSON 3 — DAY 5

Goal: Technical documentation

Create:

```
docs/
├── gaze_method.md
├── environment_method.md
├── camera_setup.md
└── technical_limitations.md
```

Include

Gaze

- MediaPipe
- landmarks
- calibration concept
- gaze features
- head pose
- FPS
- limitations

Environment

- YOLO
- object classes
- confidence

- feature extraction

Camera

- primary face camera
- independent environment camera
- geometry
- synchronization concept

Very important

Clearly distinguish:

```
Implemented/tested
```

from:

```
Proposed future deployment
```

For example:

> "CCTV replacement is proposed as a modular deployment extension; it was not experimentally validated in this five-day prototype."

That wording is much safer scientifically because the literature review specifically found that the exact online-to-CCTV substitution architecture was not verified in prior work but also recommended formal validation before making a strong claim.
Literature_Review_and_Novelty_A…

PERSON 4 — DAY 5

Goal: Final QA + figures + reproducibility

Step 1

Run the entire pipeline from scratch.

```bash
python generate_dataset.py
python train.py
python run_experiments.py
python generate_figures.py
```

Step 2

Create final tables.

Table 1

Dataset statistics.

Table 2

Main model comparison.

```
Model | AUC | F1 | Precision | Recall | FPR
```

Table 3

Ablation.

Table 4

Error/fairness analysis.

Step 3

Generate final figures.

Step 4

Create README

Include:

```
Project overview
Installation
Dataset generation
Training
Evaluation
Results
Reproduction
Limitations
```

Step 5

Run code quality

```
no broken imports
no hard-coded paths
no missing files
no undocumented dependencies
```

Step 6

Final reproducibility test

A fresh machine should be able to understand how to reproduce the experiment.

---

9. DAILY TEAM SCHEDULE

Don't have four people sitting in meetings.

Use approximately:

Time	Activity
09:00–09:20	Stand-up
09:20–12:30	Deep work
12:30–13:00	Break
13:00–16:30	Deep work
16:30–17:00	Integration/check
17:00–18:00	Documentation/commits
18:00–18:15	Daily status

Stand-up format

Each person answers only:

```
1. What did I finish?
2. What am I doing today?
3. Am I blocked?
```

No long meeting.

---

10. THE DEPENDENCY STRUCTURE

This is the part I would enforce very strictly.

Bad architecture

```
Person 2
   ↓
Person 3
   ↓
Person 1
   ↓
Person 4
```

If Person 2 gets stuck, everybody stops.

Our architecture

```
                ┌──────────────┐
                │ DATA CONTRACT│
                └──────┬───────┘
                       │
       ┌───────────────┼────────────────┐
       ↓               ↓                ↓
   Person 1        Person 2         Person 3
   Fusion          Dataset          CV
       │               │                │
       └───────────────┼────────────────┘
```

```
                   ↓
                Person 4
              Evaluation
```

But even here Person 4 uses mock data initially.

So:

```
Person 1 → mock data
Person 2 → independent simulator
Person 3 → independent CV
Person 4 → mock predictions
```

Then on Day 3:

```
Person 1 ← actual dataset
Person 4 ← actual dataset
Person 3 → technical evidence
```

No one is fundamentally blocked.

---

11. WHAT EACH PERSON MUST HAND OVER

This prevents the classic team problem where someone says:

> "My part is basically done."

No.

Each person has a **defined acceptance test**.

PERSON 1 ACCEPTANCE TEST

Must have:

```
[✓] Early fusion
[✓] Late fusion
[✓] Attention fusion
[✓] Single-stream models
[✓] Ablation models
[✓] Explainable scoring
[✓] Model save/load
[✓] Training script
[✓] Methodology text
[✓] Paper draft
```

PERSON 2 ACCEPTANCE TEST

```
[✓] 200 sessions
[✓] Honest + cheating
[✓] Multiple cheating types
[✓] 10-second windows
[✓] Correct schema
[✓] No NaNs
[✓] No leakage
```

```
[✓] Reproducible seed
[✓] Dataset documentation
```

PERSON 3 ACCEPTANCE TEST

```
[✓] Gaze works
[✓] Head pose works
[✓] Gaze features extracted
[✓] YOLO works
[✓] Environment features extracted
[✓] FPS measured
[✓] Example outputs saved
[✓] Technical limitations documented
```

PERSON 4 ACCEPTANCE TEST

```
[✓] AUC
[✓] F1
[✓] Precision
[✓] Recall
[✓] FPR
[✓] FNR
[✓] ROC
[✓] PR
[✓] Confusion matrix
[✓] Ablation
[✓] Error analysis
[✓] Final tables
[✓] README
[✓] Reproduction test
```

12. EXPERIMENTS YOU ABSOLUTELY SHOULD NOT SKIP

If you're short on time, **do not add more features**. Do these experiments.

Experiment 1 — Single modality

```
Gaze only
Mouse only
Environment only
```

Question:

> How much can each modality detect independently?

Experiment 2 — Fusion comparison

```
Early
Late
Attention
```

Question:

> Does the proposed fusion architecture improve over simpler fusion?

This is central to your contribution.

Experiment 3 — Ablation

```
Gaze + Mouse
Gaze + Environment
Mouse + Environment
Gaze + Mouse + Environment
```

Question:

> Does every stream contribute?

Experiment 4 — Error analysis

Study:

```
False positives
False negatives
```

Especially:

```
normal looking away
high mouse activity
accessibility-like behavior
temporary tab changes
occlusion
poor gaze estimation
```

Experiment 5 — Robustness

At minimum:

```
gaze noise
missing environment stream
noisy mouse
single-modality cheating
```

This is far more valuable than building another fancy model.

---

13. IMPORTANT CHANGE TO THE EXISTING PLAN: DON'T PROMISE SPECIFIC RESULTS

The existing sprint contains illustrative results such as:

> Attention Fusion = 0.85 AUC

and expected ordering such as:

> Attention > Late > Early.

ONE_WEEK_SPRINT_PLAN

Do not put those numbers into your final paper unless your actual experiment produces them.

Instead, write the hypothesis beforehand:

> **H1:** Three-stream attention fusion will outperform single-modality models.
>
> **H2:** Attention fusion will outperform early and late fusion baselines.
>
> **H3:** Removing any one modality will reduce performance.

Then run the experiment.

If H1/H2/H3 fail, **that is still a research result**.

---

14. WHAT YOU SHOULD NOT TRY TO DO IN THESE 5 DAYS

This is extremely important.

Don't attempt:

❌ Real large-scale participant collection

You don't have time.

The complete plan assumes months of work for real data collection, ethics, deployment and validation.

Complete_Plan

❌ Train a large gaze model

Use MediaPipe/pretrained methods.

❌ Train YOLO from scratch

Use pretrained YOLO.

❌ Build full CCTV deployment

Only design the architecture.

❌ Build a production proctoring platform

You're building a research prototype.

❌ Claim "first multimodal proctoring system"

Your literature review explicitly says that would be indefensible.

Literature_Review_and_Novelty_A…

❌ Claim "AI detects cheating with certainty"

Your output should be:

```
risk/incident score
+
evidence
+
human review
```

not:

```
CHEATING = TRUE
```

15. FINAL PAPER POSITIONING

I strongly recommend positioning the paper around:

"Explainable Three-Stream Fusion for Exam Malpractice Detection"

rather than:

> "A new eye-tracking system"

or:

> "A new cheating detection camera"

or:

> "The first multimodal proctoring system."

Core contribution

You have three established sensing channels:

```
Gaze
   +
Interaction
   +
Environment
```

and the research contribution becomes:

```
Synchronized
      ↓
Feature extraction
      ↓
Stream-specific representation
      ↓
Cross-stream attention
      ↓
Learned stream weighting
      ↓
Calibrated risk score
```

```
       ↓
Human-reviewable explanation
```

This aligns directly with the novelty assessment, which recommends making the **defined three-stream fusion model**, synchronization, explainable calibrated scoring, and ablation evidence the central contribution.
Literature_Review_and_Novelty_A…

16. YOUR FINAL 5-DAY TIMELINE

DAY 1 — FREEZE

Person 1
→ architecture + schema + fusion skeleton

Person 2
→ simulator architecture

Person 3
→ gaze + YOLO skeleton

Person 4
→ evaluation framework

End of Day 1

```
Architecture ✓
Schema ✓
Mock data ✓
Simulator skeleton ✓
CV skeleton ✓
Evaluation skeleton ✓
```

DAY 2 — BUILD

Person 1
→ fusion models

Person 2
→ complete simulator + dataset

Person 3
→ CV extraction

Person 4
→ training/evaluation pipeline

End of Day 2

```
All components independently executable ✓
```

DAY 3 — INTEGRATE

Person 1
→ train fusion models

Person 2
→ validate dataset

Person 3
→ validate CV + collect technical measurements

Person 4
→ first complete experiment

End of Day 3

```
First end-to-end result ✓
```

DAY 4 — PROVE

Person 1
→ proposed model + explainability

Person 2
→ robustness/noise datasets

Person 3
→ architecture figures + CV evidence

Person 4
→ all experiments + statistics + figures

End of Day 4

```
Final experimental results ✓
```

DAY 5 — PACKAGE

Person 1
→ paper

Person 2
→ dataset documentation

Person 3
→ technical documentation

Person 4
→ QA + results + README

End of Day 5

```
CODE ✓
DATA ✓
MODEL ✓
EXPERIMENTS ✓
FIGURES ✓
PAPER ✓
README ✓
```

17. THE ONE THING I WOULD ADD THAT THE ORIGINAL PLAN IS MISSING

Create a **"Claim–Evidence Matrix"** on Day 1.

File:

```
docs/claim_evidence_matrix.md
```

Example:

Claim	Evidence required	Owner
Three streams improve detection	Ablation	P4
Attention beats simple fusion	Model comparison	P1/P4
Gaze provides useful signal	Gaze-only baseline	P4
Mouse provides useful signal	Mouse-only baseline	P4
Environment provides useful signal	Env-only baseline	P4
System is explainable	Attention/gating examples	P1
Gaze runs in real time	FPS measurement	P3
Environment detection works	Detection examples	P3
Dataset is reproducible	Generator + seed	P2
CCTV extension is possible	Architecture	P1/P3
CCTV extension is experimentally validated	**NOT CLAIMED**	—

This will prevent the paper from making claims that the experiments don't support.

---

18. THE FINAL TEAM ORGANIZATION IN ONE TABLE

	Person 1	Person 2	Person 3	Person 4
Primary role	Research/Fusion	Data	Computer Vision	Evaluation
Main ownership	Model + paper	Dataset	Gaze + environment	Experiments
Day 1	Architecture	Simulator	CV	Metrics
Day 2	Fusion	Data generation	CV implementation	Training
Day 3	Integration	Data QA	CV validation	First results
Day 4	Proposed model	Robustness data	Figures/technical evidence	Full experiments
Day 5	Paper	Dataset docs	Technical docs	QA/results

	Person 1	Person 2	Person 3	Person 4
Main output	Paper + fusion	Dataset	CV modules	Results
Can work with mock data?	Yes	Yes	Yes	Yes
Can work independently?	**Yes**	**Yes**	**Yes**	**Yes**
Final responsibility	Scientific validity	Data validity	Sensor validity	Experimental validity

Bottom line

Do not think of this as four people completing four sequential pieces.

Think of it as:

> **Four independent pipelines converging on one fixed interface.**

That is what makes a 5-day deadline realistic.

And I would make **Day 3 the hard integration deadline**: if you don't have at least one complete end-to-end result by the end of Day 3, stop adding features and spend Days 4–5 on making the existing experiment scientifically solid.

Your existing project documents support exactly this scope: the one-week version is intentionally a **simulation-based prototype**, with the novelty concentrated in the **fusion architecture**, while real exams, real CCTV deployment, large-scale datasets and fairness validation remain future work.

ONE_WEEK_SPRINT_PLAN

If you follow the allocation above, **no single person becomes a bottleneck**, and each person's work has a concrete definition of "done."

Sources